

Calculus and Linear Algebra

Functions and Their Graphs

with a short closure on inverse matrices

Lecture 10

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Lecture outline

- ▶ Closing the Linear Algebra unit: inverse matrices.
- ▶ What a function is: domain, codomain, image and graph.
- ▶ Vertical line test, zeros and reading information from a graph.
- ▶ Monotonicity, injectivity, surjectivity, bijectivity.
- ▶ Inverse functions and composition.
- ▶ Even and odd functions; transformations of graphs.

Main goal

Start building the language of Calculus by learning how to **read**, **transform** and **compare** functions.

Closing the Linear Algebra unit: inverse matrix

For a square matrix A , an **inverse matrix** is a matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I.$$

Meaning

Multiplying by A^{-1} undoes the action of A , just as multiplying by $\frac{1}{a}$ undoes multiplication by $a \neq 0$.

- ▶ Only **square** matrices can have an inverse.
- ▶ If A^{-1} exists, then it is **unique**.
- ▶ A square matrix is invertible exactly when $\det(A) \neq 0$.

How do we compute A^{-1} ?

Two standard methods:

1. **Shortcut for 2×2 matrices**

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad (ad - bc \neq 0).$$

2. **Gauss–Jordan method** for any size: start from the augmented matrix

$$[A \mid I]$$

and row-reduce until the left block becomes I . The right block becomes A^{-1} .

Example: computing an inverse

Exercise Find the inverse of

$$A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}.$$

First check invertibility:

$$\det(A) = 2 \cdot 3 - 1 \cdot 5 = 1 \neq 0.$$

Thus A is invertible. Using the 2×2 formula,

$$A^{-1} = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}.$$

Same answer by Gauss–Jordan

Reducing $\begin{bmatrix} 2 & 1 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{bmatrix}$ also gives $\begin{bmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -5 & 2 \end{bmatrix}$.

Check: matrix times inverse gives the identity

$$\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

Also,

$$\begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Why this is useful

If $Ax = b$ and A is invertible, then multiplying by A^{-1} gives

$$x = A^{-1}b.$$

Transition: from matrices to functions

Linear systems study relationships among several unknowns. Calculus starts with a different basic object:

one input $\mapsto$ one output.

Examples:

$$x \mapsto x^2, \quad x \mapsto \sqrt{x}, \quad x \mapsto \frac{1}{x-2}.$$

Today

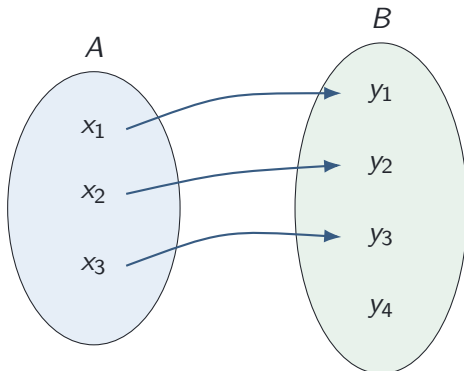
We begin by understanding what a function is, how its graph encodes information, and how formulas change the graph.

What is a function?

A function f from a set A to a set B is a rule that assigns

to each $x \in A$ exactly one value $f(x) \in B$.

- ▶ A is the **domain**.
- ▶ B is the **codomain**.
- ▶ The set of all actual outputs is the **image** or **range**.
- ▶ Notation: $f : A \rightarrow B$.



One input, one output

The key condition is **uniqueness of the output**.

Function

Each input has exactly one output.

allowed: $x \mapsto x^2$

Not a function

One input giving two outputs is not allowed.

not allowed: $x \mapsto \pm\sqrt{x}$

Important

Different inputs are allowed to have the same output. The restriction is only: *one input cannot produce two different outputs*.

Domain, codomain and image: an example

Consider

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

- ▶ **Domain:** $\mathbb{R}$.
- ▶ **Codomain:** $\mathbb{R}$.
- ▶ **Image:** $[0, +\infty)$.

Observation

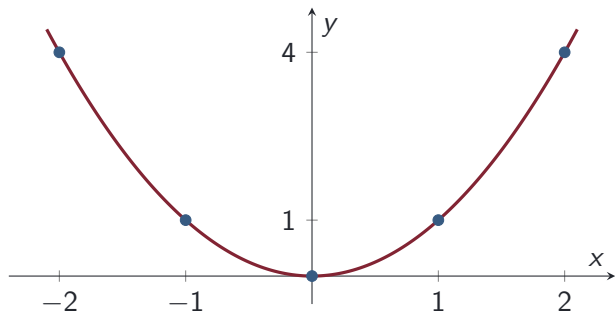
The codomain is declared in the definition; the image is the set of values that the function actually reaches.

- ▶ $f(-2) = 4$, $f(0) = 0$, $f(3) = 9$.
- ▶ The value -1 belongs to the codomain, but not to the image.

The graph of a function

For a real function $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$, its graph is the set of points

$$\{(x, f(x)) : x \in D\}.$$

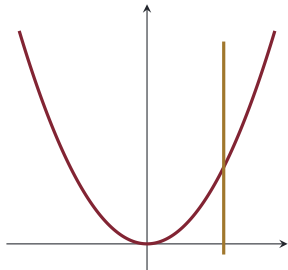


Each point on the graph tells us: if the input is x , then the output is $y = f(x)$.

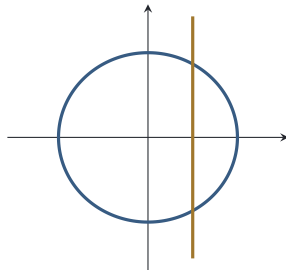
Vertical line test

A curve in the plane is the graph of a function of x **iff** every vertical line meets it at most once.

Function



Not a function



A vertical line cuts the circle in two points, so the circle is not the graph of a function of x .

Reading information from a graph

From the graph we can identify:

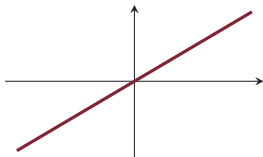
- ▶ **function values:** the height $y = f(x)$ at a given x ;
- ▶ **zeros:** points where the graph meets the x -axis;
- ▶ **y -intercept:** the value $f(0)$ when 0 is in the domain;
- ▶ **sign:** where the graph is above or below the x -axis;
- ▶ **qualitative behaviour:** increasing, decreasing, symmetric, bounded, etc.

Later on

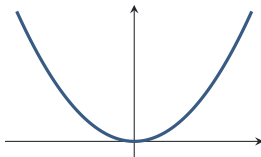
Limits, derivatives and complete function study all start from careful graph reading.

Some basic families of functions

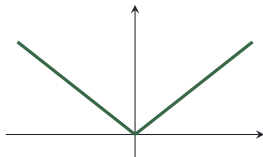
Linear $y = x$



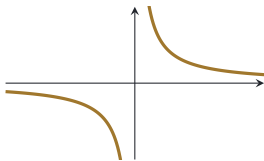
Quadratic $y = x^2$



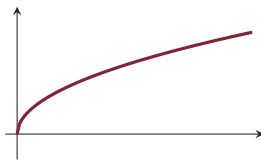
Absolute value $y = |x|$



Reciprocal $y = 1/x$



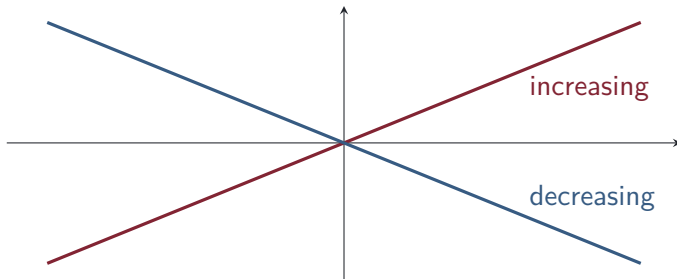
Square root $y = \sqrt{x}$



Different formulas produce very different domains and shapes.

Monotonicity: the geometric idea

A function is **increasing** if larger inputs give larger outputs. A function is **decreasing** if larger inputs give smaller outputs.



Monotonicity is a first way of describing the global behaviour of a graph.

Monotonicity: formal definition

Let I be an interval.

► f is **increasing** on I if

$$x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2).$$

► f is **strictly increasing** on I if

$$x_1 < x_2 \Rightarrow f(x_1) < f(x_2).$$

► f is **decreasing** on I if

$$x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2).$$

► f is **strictly decreasing** on I if

$$x_1 < x_2 \Rightarrow f(x_1) > f(x_2).$$

Examples

x^3 is strictly increasing on $\mathbb{R}$; x^2 is decreasing on $(-\infty, 0]$ and increasing on $[0, +\infty)$.

Injective functions

A function $f : A \rightarrow B$ is **injective** if different inputs always produce different outputs:

$$f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

Equivalent form:

$$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2).$$

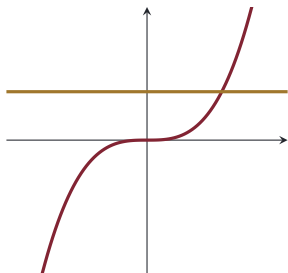
Graphical test

A real function is injective on an interval iff every horizontal line meets the graph at most once.

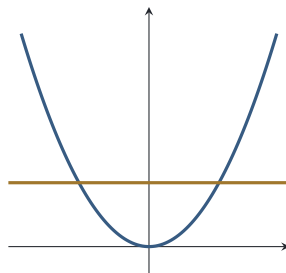
Strictly monotone functions are injective.

Horizontal line test

Injective



Not injective



The parabola fails the test because the same output may come from two different inputs.

Surjective and bijective functions

Let $f : A \rightarrow B$.

- ▶ f is **surjective onto** B if every $y \in B$ is hit by at least one $x \in A$.

$$\text{Im}(f) = B.$$

- ▶ f is **bijective** if it is both injective and surjective.

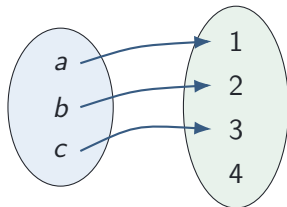
Why bijectivity matters

Only bijective functions admit a genuine inverse function from B back to A .

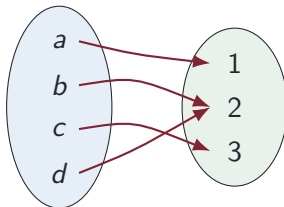
Function	As map $\mathbb{R} \rightarrow \mathbb{R}$	Comment
$f(x) = x$	bijective	one-to-one and onto
$f(x) = x^2$	not injective, not surjective	misses negative values
$f(x) = e^x$	injective, not surjective	image $(0, +\infty)$

Finite-set pictures: injective, surjective, bijective

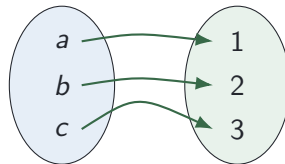
injective



surjective



bijective



Inverse function: the idea

If $f : A \rightarrow B$ is bijective, the inverse function

$$f^{-1} : B \rightarrow A$$

reverses the correspondence:

$$f^{-1}(f(x)) = x, \quad f(f^{-1}(y)) = y.$$

Key fact

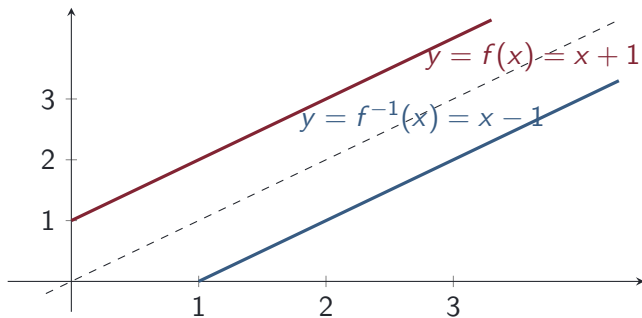
A function has an inverse function exactly when it is bijective.

The roles of domain and image are exchanged:

$$\text{Dom}(f^{-1}) = \text{Im}(f), \quad \text{Im}(f^{-1}) = \text{Dom}(f).$$

Graph of the inverse: reflection across $y = x$

If f is invertible, then the graph of f^{-1} is the reflection of the graph of f across the line $y = x$.



Points (a, b) on the graph of f become points (b, a) on the graph of f^{-1} .

How to compute an inverse function

Exercise Find the inverse of

$$f(x) = 2x - 3.$$

Solution Set $y = 2x - 3$ and solve for x :

$$y = 2x - 3 \implies 2x = y + 3 \implies x = \frac{y + 3}{2}.$$

Now rename the variable:

$$f^{-1}(x) = \frac{x + 3}{2}.$$

Check

$$f(f^{-1}(x)) = 2 \cdot \frac{x + 3}{2} - 3 = x, \quad f^{-1}(f(x)) = \frac{2x - 3 + 3}{2} = x.$$

Why x^2 is not invertible on all of $\mathbb{R}$

For

$$f(x) = x^2,$$

we have

$$f(-1) = f(1) = 1.$$

So f is not injective on $\mathbb{R}$, hence it cannot have an inverse function on all of $\mathbb{R}$. However, if we restrict the domain to $[0, +\infty)$, then f becomes injective and

$$f^{-1}(x) = \sqrt{x}, \quad x \geq 0.$$

Lesson

Sometimes a function becomes invertible only after restricting its domain.

Composition of functions

Given

$$f : A \rightarrow B, \quad g : B \rightarrow C,$$

the composition $g \circ f$ is defined by

$$(g \circ f)(x) = g(f(x)).$$

Meaning

First apply f , then apply g to the result.

$$A \xrightarrow{f} B \xrightarrow{g} C.$$

Composition is one of the most important operations in Calculus.

Composition: example and domain check

Let

$$f(x) = x + 1, \quad g(x) = x^2.$$

Then

$$(g \circ f)(x) = g(x + 1) = (x + 1)^2,$$

while

$$(f \circ g)(x) = f(x^2) = x^2 + 1.$$

Important

In general,

$$g \circ f \neq f \circ g.$$

If one of the functions has a restricted domain, we must also check that the output of the first function lies in the domain of the second.

A domain issue in composition

Let

$$f(x) = x - 2, \quad g(x) = \sqrt{x}.$$

Then

$$(g \circ f)(x) = \sqrt{x - 2},$$

which is defined only for

$$x - 2 \geq 0 \iff x \geq 2.$$

On the other hand,

$$(f \circ g)(x) = \sqrt{x} - 2,$$

which is defined for

$$x \geq 0.$$

Take care

The formula and the domain both matter.

Even and odd functions

Let D be a domain symmetric with respect to 0.

► f is **even** if

$$f(-x) = f(x) \quad \text{for all } x \in D.$$

► f is **odd** if

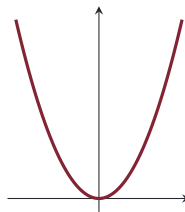
$$f(-x) = -f(x) \quad \text{for all } x \in D.$$

Geometric meaning

Even functions are symmetric with respect to the y -axis. Odd functions are symmetric with respect to the origin.

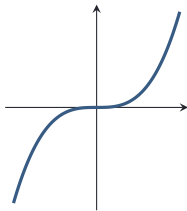
Examples of even, odd and neither

even



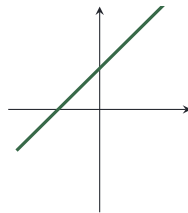
$$f(x) = x^2$$

odd



$$f(x) = x^3$$

neither



$$f(x) = x + 1$$

Quick consequences of parity

- ▶ If f is even, it is enough to study the graph for $x \geq 0$ and reflect it across the y -axis.
- ▶ If f is odd, it is enough to study the graph for $x \geq 0$ and then apply symmetry through the origin.
- ▶ The sum of two even functions is even.
- ▶ The sum of two odd functions is odd.
- ▶ The product of two even functions is even.
- ▶ The product of two odd functions is even.
- ▶ The product of an even and an odd function is odd.

Useful examples

$\cos x$ is even, $\sin x$ is odd.

Transformations of graphs: the big picture

Starting from a known graph $y = f(x)$, many other graphs can be obtained by simple transformations.

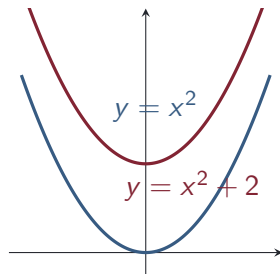
Formula	Effect on the graph
$y = f(x) + c$	vertical shift
$y = f(x - a)$	horizontal shift
$y = af(x)$	vertical stretch/compression
$y = f(bx)$	horizontal compression/stretch
$y = -f(x)$	reflection across the x -axis
$y = f(-x)$	reflection across the y -axis

We always compare the new graph with a familiar **parent function**.

Vertical shifts: $y = f(x) + c$

- ▶ If $c > 0$, the graph shifts **up** by c units.
- ▶ If $c < 0$, the graph shifts **down** by $|c|$ units.

Example: $y = x^2 + 2$



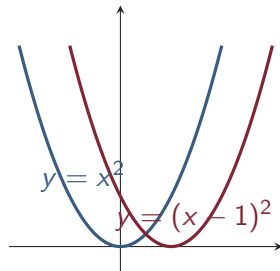
Horizontal shifts: $y = f(x - a)$

- ▶ If $a > 0$, the graph shifts **right** by a units.
- ▶ If $a < 0$, the graph shifts **left** by $|a|$ units.

Example: $y = (x - 1)^2$

Common mistake

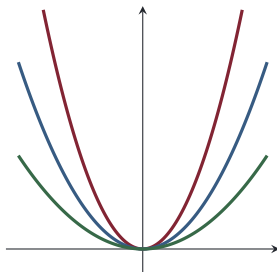
Inside the brackets, the direction is reversed:
 $x - 1$ means “shift right by 1”.



Vertical stretching and compression: $y = af(x)$

- ▶ If $|a| > 1$, the graph is stretched away from the x-axis.
- ▶ If $0 < |a| < 1$, the graph is compressed toward the x-axis.
- ▶ If $a < 0$, there is also a reflection across the x-axis.

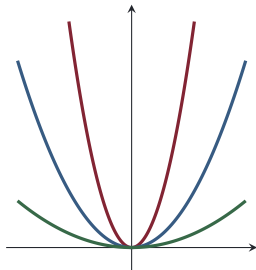
Example: $y = 2x^2$, $y = \frac{1}{2}x^2$



Horizontal stretching and compression: $y = f(bx)$

- ▶ If $|b| > 1$, the graph is **compressed horizontally**.
- ▶ If $0 < |b| < 1$, the graph is **stretched horizontally**.

Example: $y = (2x)^2 = 4x^2$, $y = \left(\frac{x}{2}\right)^2$



Again: inside means reverse intuition

Multiplying x by 2 makes the graph narrower, not wider.

Reflections

Across the x -axis

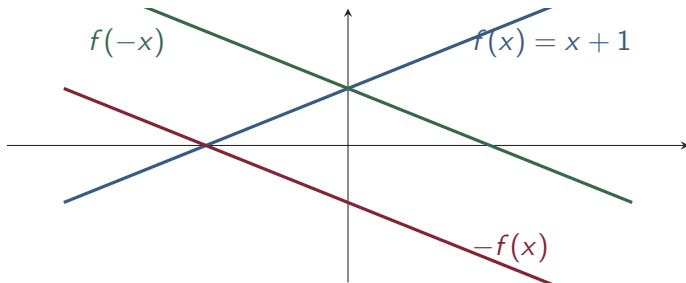
$$y = -f(x)$$

Every y -value changes sign.

Across the y -axis

$$y = f(-x)$$

Every x -value changes sign.



Building a graph step by step

Start with the parent function

$$f(x) = |x|.$$

Now consider

$$g(x) = 2|x - 1| - 3.$$

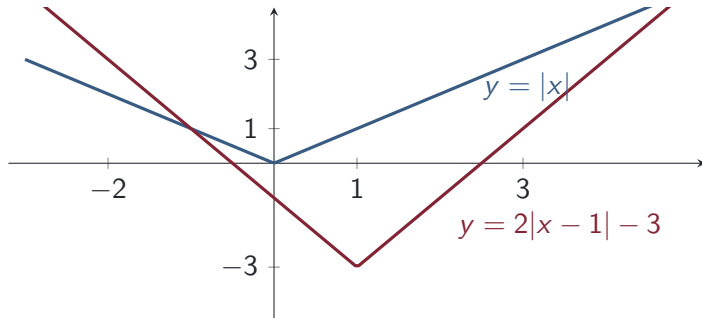
Read the formula from inside to outside:

1. $x - 1$: shift right by 1.
2. $|\cdot|$: keep the V-shape.
3. $2\cdot$: vertical stretch by factor 2.
4. -3 : shift down by 3.

Method

To sketch a transformed graph, identify the parent function first, then apply one transformation at a time.

Example: parent graph and transformed graph



The vertex moves from $(0, 0)$ to $(1, -3)$, and the arms become steeper.

Worked exercise: classify key properties

Exercise For each function, decide whether it is injective, surjective as a map $\mathbb{R} \rightarrow \mathbb{R}$, even, odd, or neither.

$$\text{A: } f(x) = x^3, \quad \text{B: } g(x) = x^2, \quad \text{C: } h(x) = x + 1.$$

Hints

Use the graph, the horizontal line test, and the definitions of even and odd functions.

Worked exercise: solutions

Solution

- ▶ $f(x) = x^3$: injective, surjective, hence bijective; odd.
- ▶ $g(x) = x^2$: not injective, not surjective as a map $\mathbb{R} \rightarrow \mathbb{R}$; even.
- ▶ $h(x) = x + 1$: injective and surjective, hence bijective; neither even nor odd.

Why?

x^3 is strictly increasing; x^2 fails the horizontal line test and has image $[0, +\infty)$; $x + 1$ is a line with slope 1 and intercept 1.

Worked exercise: compositions and inverses

Exercise Let

$$f(x) = 2x - 3, \quad g(x) = x^2.$$

Compute:

$$(g \circ f)(x), \quad (f \circ g)(x), \quad f^{-1}(x).$$

Solution

$$(g \circ f)(x) = (2x - 3)^2, \quad (f \circ g)(x) = 2x^2 - 3, \quad f^{-1}(x) = \frac{x + 3}{2}.$$

The two compositions are different, but f^{-1} undoes f .

What you should be able to do after this lecture

- ▶ Recognize whether a rule defines a function.
- ▶ Distinguish domain, codomain and image.
- ▶ Read function values, zeros and sign from a graph.
- ▶ Use vertical and horizontal line tests.
- ▶ Decide injectivity, surjectivity and bijectivity.
- ▶ Compute simple inverse functions and compositions.
- ▶ Identify even and odd functions.
- ▶ Sketch basic transformations of a known graph.
- ▶ Compute the inverse of a 2×2 matrix and verify it.

Independent practice

Exercise

1. Find the inverse of $\begin{pmatrix} 4 & 1 \\ 3 & 1 \end{pmatrix}$ and verify the product.
2. Decide whether each graph represents a function of x .
3. For $f(x) = x^2 - 4$, find the zeros and decide where $f(x) \geq 0$.
4. For $f(x) = \frac{x-1}{2}$, compute $f^{-1}(x)$.
5. For $f(x) = |x|$ and $g(x) = 2|x-1| - 3$, describe all transformations.
6. Compute $f \circ g$ and $g \circ f$ for $f(x) = x + 2$, $g(x) = \sqrt{x}$, including the domains.

Looking ahead

In the next lectures we will deepen the language of functions by studying:

- ▶ domains in detail;
- ▶ sign and intersections;
- ▶ limits and continuity;
- ▶ derivatives and graph interpretation;
- ▶ later on, also functions of two variables.

Big picture

Everything in Calculus starts from one central skill: understanding how a formula becomes a graph, and how a graph encodes information.